

Feature Identification Report

Purpose

Identify mobile-phone specifications relevant to price-category classification using the actual mobile_dataset.csv schema.

The dataset contains 1,019 records and 15 raw columns.

Recommended numeric features after extraction

Spec_Score; rating; RAM_GB; Storage_GB; Battery_mAh; Charging_W; Screen_Size_Inches; Resolution_Width; Resolution_Height; Refresh_Rate_Hz; Rear_Camera_MP; Front_Camera_MP; Processor_Speed_GHz; Cores.

Recommended categorical or binary features

tag; connectivity flags from sim such as 3G, 4G, 5G, VoLTE, Wi-Fi and NFC; External_Memory_Supported; Android_Version; FM_Radio.

Feature Sources and Target

Raw columns requiring parsing

processor -> processor speed and cores; 993 usable rows.

storage -> RAM and internal storage; 999 usable rows.

battery -> battery capacity and charging power; 1,013 usable rows.

display -> screen size, resolution and refresh rate; 1,018 usable rows.

camera -> rear and front camera values; 1,008 usable rows.

sim -> connectivity flags; 1,019 usable rows.

Target

Create Price_Category from price using a documented rule. price is the target source, not an input feature.

Unwanted Columns and Timing

Exclude from model inputs

img: product URL, not a phone specification.

Name: product identifier; retain for reporting, but do not use as a predictive feature.

price: use only to create Price_Category; excluding it prevents target leakage.

Raw text columns after useful values have been extracted.

When should columns be deleted?

Feature identification decides what is relevant; it does not physically delete raw data.

Review missing values and duplicates during data cleaning.

Drop or select final columns during feature preparation before train-test splitting and model training.

Keep the original CSV unchanged as a source copy.